



SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)

Constituent of Symbiosis International (Deemed University), Pune

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Subject: Programming with Java

Sem: III AY: 2026-27

Assignment No: 9

TITLE: Write a Java program to demonstrate the use of the final keyword with variables,

methods, and classes.

Problem Statement

Write a Java program to demonstrate the use of the final keyword with variables, methods, and classes.

Learning Objectives

To implement the final keyword with variables, methods, and classes.

Theory

The final keyword restricts modification in Java programs. A final variable cannot be reassigned after initialization. A final method cannot be overridden by subclasses, while a final class cannot be inherited. This improves security, prevents accidental modifications, and increases program reliability. Final is commonly used for constants and immutable classes.

1. Create a Bank Account program where account number is final and cannot be changed.

BankAccount.java

 Share 

```
1 class BankAccount {
2     final int accountNumber;
3     String accountHolder;
4     double balance;
5
6     BankAccount(int accountNumber, String accountHolder, double balance) {
7         this.accountNumber = accountNumber;
8         this.accountHolder = accountHolder;
9         this.balance = balance;
10    }
11
12    void display() {
13        System.out.println("Account Number: " + accountNumber);
14        System.out.println("Account Holder: " + accountHolder);
15        System.out.println("Balance: " + balance);
16    }
17
18    public static void main(String[] args) {
19        BankAccount b = new BankAccount(123456, "Aryan Singh Kasera", 99000);
20
21        b.display();
22    }
23 }
```

OUTPUT

Output

```
Account Number: 123456
Account Holder: Aryan Singh Kasera
Balance: 99000.0

=== Code Execution Successful ===
```

2. Create a Library Book Management program where the book ISBN is declared as final and cannot be changed once assigned. Display the book's ISBN, title, author, and price.

Book.java

 **Run**

```
1 class Book {
2     final String isbn;
3     String title;
4     String author;
5     double price;
6
7     Book(String isbn, String title, String author, double price) {
8         this.isbn = isbn;
9         this.title = title;
10        this.author = author;
11        this.price = price;
12    }
13
14    void display() {
15        System.out.println("ISBN: " + isbn);
16        System.out.println("Title: " + title);
17        System.out.println("Author: " + author);
18        System.out.println("Price: " + price);
19    }
20
21    public static void main(String[] args) {
22        Book b = new Book("982258948", "Java Programming", "Aryan Singh Kasera",
23                           900);
24
25        b.display();
26    }
```

OUTPUT

Output

```
ISBN: 982258948  
Title: Java Programming  
Author: Aryan Singh Kasera  
Price: 900.0
```

```
=== Code Execution Successful ===
```

Conclusion:

This program demonstrates the use of the `final` keyword with variables, methods, and classes in Java. It shows that `final` helps prevent modification, method overriding, and class inheritance, making programs more secure, reliable, and easier to maintain.